Topic to discuss

- → Picard Method
- → Numerical Problem
- → Homework Problem

Picard Method :

The Picard Method is an iterative technique used to find an approximate solution of a first-order ordinary differential equation (ODE).

It is based on converting the differential equation into an integral equation and then solving it step by step using iteration.

Standard Form of ODE :

Consider the differential equation :

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0$$

we can integrate this to obtain the solution in the interval (x_0, x)

$$\int_{x_0}^{x} dy = \int_{x_0}^{x} f(x, y)\,dx$$

$$\Rightarrow \quad y(x) = y(x_0) + \int_{x_0}^{x} f(x, y)\,dx$$

The iterative equation is written as

$$y_{n+1} = y_0 + \int_{x_0}^{x} f(x, y_n)\,dx$$

Numerical Problem :

Q: Solve the equation by Picard's method

$$y'(x) = x e^{y} \quad , \quad y(0) = 0$$

and estimate $y(0.1)$, $y(0.2)$ and $y(1)$

Solution: Lets considered,

$$f(x,y) = \frac{dy}{dx} = y'(x)$$

so, $f(x,y) = x e^{y}$

also, $x_0 = 0$

$y_0 = 0$

First Iteration :

Taking $n=0$, $x_0=0$, $y_0=0$, $f(x,y) = x e^{y}$

So, $y_{n+1} = y_0 + \int_{x_0}^{x} f(x, y_n)\, dx$ becomes

$$\Rightarrow y_1 = y_0 + \int_0^x f(x, y_0)\, dx$$

$$\Rightarrow y_1 = 0 + \int_0^x (x\, e^{y_0})\, dx$$

$$= 0 + \int_0^x (x \cdot e^0)\, dx = \int_0^x x\, dx$$

$$\therefore y_1 = \frac{x^2}{2}$$

Second Iteration :

Taking $n = 1$, $x_0 = 0$, $y_0 = 0$, $y_1 = \frac{x^2}{2}$, $f(x, y) = x e^y$

So, $y_{n+1} = y_0 + \int_{x_0}^{x} f(x, y_n)\, dx$

$$\Rightarrow y_2 = 0 + \int_0^x f(x, y_1)\, dx$$

$$= \int_0^x x\, e^{y_1}\, dx = \int_0^x x \times e^{\frac{x^2}{2}}\, dx$$

$$= e^{x^2/2} - 1$$

$\int_0^x x\, e^{x^2/2}\, dx = ?$

let $u = \frac{x^2}{2} \Rightarrow du = x\,dx$

Replace in integral,

$\int e^{x^2/2} \cdot x\,dx = \int e^u\, du = e^u + C$

substitute,

$\int_0^x e^{x^2/2} x\,dx = \left[e^{x^2/2}\right]_0^x$

$= e^{x^2/2} - e^0$

$= (e^{x^2/2} - 1)$ Ans.

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3^{rd} iteration :

Taking $n=2$, $x_0=0$, $y_0=0$, $y_2=e^{x^2/2}-1$, $f(x,y)=xe^y$

So, $y_{n+1}=y_0+\int_{x_0}^{x} f(x,y_n)\,dx$

$$\Rightarrow y_3=y_0+\int_{x_0}^{x} f(x,y_2)\,dx$$

$$y_3=0+\int_0^x xe^{y_2}\,dx$$

$$y_3=\int_0^x xe^{\left(e^{x^2/2}-1\right)}\,dx$$

This integration is not so easy to solve directly

We need to expand this to solve,

So at this point have,

$$y_2 = e^{x^2/2} - 1$$

So, $y(x) = e^{x^2/2} - 1$

we also have to find $y(0.1)$, $y(0.2)$ and $y(1)$

So, $y(0.1) = e^{0.1^2/2} - 1 = 0.00501252$

$y(0.2) = e^{0.2^2/2} - 1 = 0.020201$

$y(1) = e^{1^2/2} - 1 = 0.648721$

* Write only If asked in Exam

we know, exact solution of $y'(x) = xe^y$ is

$$\frac{dy}{dx} = xe^y$$

$$\Rightarrow \frac{dy}{e^y} = x\,dx$$

$$\Rightarrow e^{-y}\,dy = x\,dx$$

$$\Rightarrow \int e^{-y}\,dy = \int x\,dx$$

$$\Rightarrow -e^{y} = \frac{x^2}{2} + C$$

$$\Rightarrow -e^{y} = \frac{x^2}{2} - 1$$

Use initial condition,

$y(0) = 0$

Put $x = 0$ $y = 0$

$\therefore -e^0 = 0 + C$

$\therefore C = -1$

$$\Rightarrow -e^{y} = \frac{x^2}{2} - 1$$

$$\Rightarrow e^{-y} = 1 - \frac{x^2}{2}$$

Take log,

$$-y = \ln\left(1 - \frac{x^2}{2}\right)$$

$$\therefore y = -\ln\left(1 - \frac{x^2}{2}\right)$$

Therefore,

$$y(0.1)_{exact} = 0.0050125$$

$$y(0.2)_{exact} = 0.0202027$$

$$y(1)_{exact} = 0.6933147$$

Homework Problem :

Q: Solve $y'(x) = x^2 + y^2$, $y(0) = 0$ using Picard Method, and also estimate $y(0.1)$ and $y(0.2)$

Solution: Lets considered,

$$f(x,y) = \frac{dy}{dx} = y'(x)$$

so, $$f(x,y) = x^2 + y^2$$

also, $x_0 = 0$

$y_0 = 0$

first Iteration :

Taking, $n=0$, $x_0=0$, $y_0=0$, $f(x,y) = x^2+y^2$

So, $y_{n+1} = y_0 + \int_{x_0}^{x} f(x, y_n)\, dx$ becomes,

$$y_1 = y_0 + \int_{x_0}^{x} f(x, y_0)\, dx$$

$$= 0 + \int_{0}^{x} (x^2 + y_0^2)\, dx$$

$$= \int_{0}^{x} (x^2+0)\, dx = \int_{0}^{x} x^2\, dx = \frac{x^3}{3}$$

$$\therefore y_1 = \frac{x^3}{3}$$

Second Iteration :

Taking, $n=1$, $x_0=0$, $y_0=0$, $y_1=\frac{x^3}{3}$, $f(x,y)=x^2+y^2$

So, $y_{n+1}=y_0+\int_{x_0}^{x} f(x,y_n)\,dx$ becomes,

$$\Rightarrow y_2 = 0+\int_0^x f(x,y_1)\,dx$$

$$\Rightarrow y_2 = \int_0^x (x^2+y_1^2)\,dx = \int_0^x \left(x^2+\frac{x^6}{9}\right)dx$$

$$\Rightarrow y_2 = \frac{x^3}{3}+\frac{x^7}{63}$$

This process can be continued further although it may be a difficult task.

If we stop at. y_2 then,

$$y(x) = \frac{x^3}{3} + \frac{x^7}{63}$$

Therefore,

$$y(0.1) = \frac{0.1^3}{3} + \frac{0.1^7}{63} = 0.000333$$

$$y(0.2) = \frac{0.2^3}{3} + \frac{0.2^7}{63} = 0.00266687$$

Ans.

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